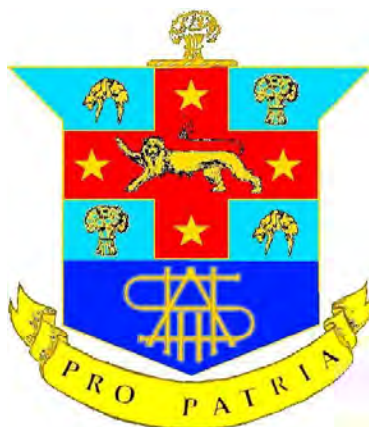




Physics

HSC Course

2012

Year 12 Trial HSC Examination

Total marks 100

General Instructions

- Reading time – 5 minutes
- Working time – 3 hours
- Attempt all questions
- Write using blue or black pen
- Draw diagrams using pencil
- Approved calculators may be used
- Write your I.D. number on each answer sheet
- Liquid paper must NOT be used on this paper
- For your convenience, the multiple choice answer sheet and the data sheets at the back may be removed from the rest of the paper

Total marks – 100

Section I

75 marks

This section has two parts, Part A and Part B

Part A – 20 marks

Attempt questions 1-20 (multiple choice)

Allow about 35 minutes for this part.

Part B – 55 marks

Attempt questions 20-33

Allow about 1 hour and 40 minutes for this part

Section II – Medical Physics Module

25 marks

Attempt question 34

Allow about 45 minutes for this part.

Answer this section in a separate booklet.

Teachers: Mr Coombes, Mr Robson, Mr Trotter

Task Weighting: 40 %

Multiple-choice Answer Sheet

Select the alternative A, B, C or D that best answers the question. Fill in the response oval completely.

Sample $2 + 4 =$ (A) 2 (B) 6 (C) 8 (D) 9

(A) ☐(B) ☒(C) ☐(D) ☐

If you think you have made a mistake, put a cross through the incorrect answer and fill in the new answer.

(A) ☒(B) ☒(C) ☐(D) ☐

If you change your mind and have crossed out what you consider to be the correct answer, then indicate this by writing the word correct and drawing an arrow as follows:

correct
↓

(A) ☒ (B) ☒ (C) ☐ (D) ☐

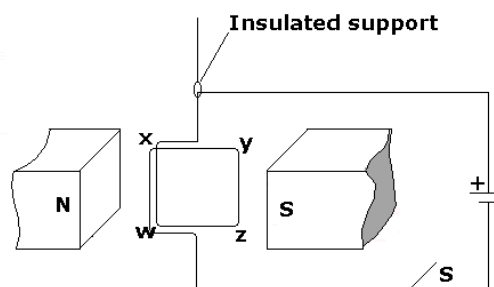
Section I – Part A

- Which statement is correct about a satellite in orbit around the Earth?
 - The gravitational acceleration is zero.
 - The gravitational acceleration of the satellite is equal to 9.8ms^{-2} .
 - The direction of gravitational acceleration is perpendicular to the satellite's velocity.
 - The gravitational force acting on the satellite is independent of the mass of the satellite.
- Three identical moons, X, Y and Z orbit the same planet. The moons have identical orbital speeds and masses of M, 9M and 25M respectively. Which response correctly shows the ratio of the orbital radii of the moons?

	Moon X	Moon Y	Moon Z
A	R	R	R
B	R	3R	5R
C	R	9R	25R
D	R	R/3	R/5

- If air resistance is neglected, which of the following statements about projectile motion is correct?
 - v_x and v_y remain constant
 - v_x and a_y remain constant
 - v_x is constant and a_y is zero
 - v_y may change direction and a_y must change direction

4. A coil, wxyz, was connected to a battery and suspended in a magnetic field between two magnets as shown. The coil is free to move in any direction or rotate.



Which of the following statements describes the response of the coil when the switch S connecting the coil to a DC voltage supply is closed?

- (A) The entire coil moves into the page.
- (B) The whole coil moves out of the page
- (C) The coil rotates so that YZ moves out of the page
- (D) The coil rotates so that XW moves out of the page

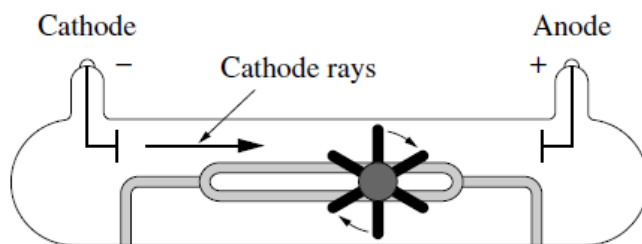
Questions 5 and 6 refer to the following experiment.

A student conducted an investigation to determine the relationship between variables A and B. The student altered variable A and then measured the corresponding changes in variable B. This was repeated three times for each magnitude of variable A. The following table was constructed from the investigation.

Variable A	Variable B (1)	Variable B (2)	Variable B (3)	Variable B (average)
5	7.4	7.5	7.3	7.4
10	30.3	30.2	30.4	30.3
15	67.4	67.6	67.5	67.5
20	121	122	120	121

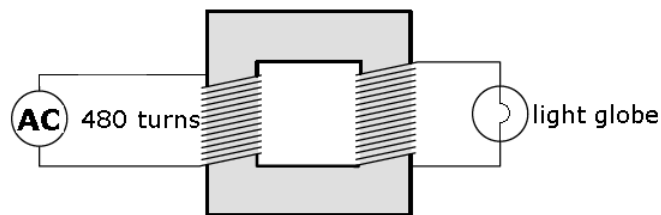
5. What do the results indicate about the relationship between variables A and B?
- (A) Variable B is the dependant variable and the measurements in the table are valid
 - (B) Variable B is the dependant variable and the measurements in the table are reliable
 - (C) Variable B is the independant variable and the measurements in the table are valid
 - (D) Variable B is the independant variable and the measurements in the table are reliable
6. The results in the table indicate that the relationship between the variables could be
- (A) Variable A is directly proportional to variable B.
 - (B) Variable A is inversely proportional to variable B.
 - (C) Variable A is directly proportional to variable B squared.
 - (D) Variable B is directly proportional to variable A squared.

7. Which of the following statements best describes the function of the laminated iron core used in a transformer?
- (A) It minimises eddy currents and increases the magnetic flux.
 - (B) It insulates the primary coil from the secondary coil.
 - (C) It absorbs heat produced by the coils and intensifies the primary coil's field.
 - (D) It maintains a balance between the input and output voltages.
8. What assumption did Planck make that enabled him to make predictions consistent with the radiation curves produced by hot objects?
- (A) Matter is converted into energy according to the equation $E = mc^2$.
 - (B) Radiation emitted and absorbed by black body radiators is quantised.
 - (C) Radiation is emitted and absorbed as waves with energy proportional to their frequencies.
 - (D) As the temperature of a black body increases, the total energy radiated increases and the wavelength at which most energy is emitted decreases.
9. Which of the following statements is not an inference that can be made using a Paddle Wheel tube?



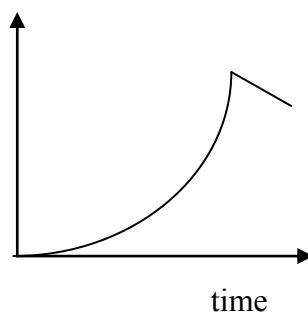
- (A) Cathode rays possess energy.
- (B) Cathode rays have momentum.
- (C) Cathode rays travel in straight lines.
- (D) Cathode rays have a negative electric charge.

10. This diagram shows a transformer that converts 240 volts to 12 V with a maximum secondary current of 10 A. The current through the 24 W 12 V light globe connected to the secondary coil is 2 A.



Four identical light globes are connected to the secondary coil. Each light globe has the same power as the original globe. Which statement is correct if the input voltage is 240V?

- (A) The number of turns on the secondary would have to be increased to 96 turns.
 - (B) The current through the primary coil of the transformer would be greater.
 - (C) The current through the secondary coil would not change.
 - (D) The voltage output of the secondary coil would be 48 volts.
11. The following graph demonstrates the motion of a rocket for the first 2 minutes after launch.



What variable should be placed on the vertical axis of this graph?

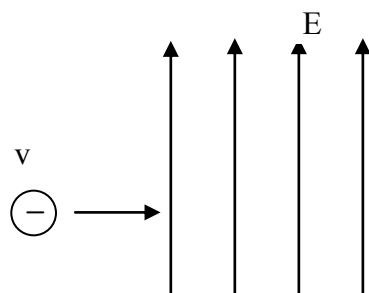
- (A) Displacement
 - (B) Velocity
 - (C) Acceleration
 - (D) G-force
12. A planet has a larger mass than the Earth but has a lower gravitational acceleration at the surface. Which combination from the table best predicts the planets mass and radius compared to Earth.

	Mass (M)	Radius (R)
A	$2M_E$	$\frac{1}{2} R_E$
B	$2 M_E$	$2 R_E$
C	$4 M_E$	$\frac{1}{2} R_E$
D	$4M_E$	$2 R_E$

13. Which of the following statements related to g-forces is most correct?

- (A) Astronauts have different tolerances for g-force but not other forces.
- (B) G-forces are stronger during launch because the craft's acceleration is positive.
- (C) Different astronauts experience different g-forces because of their different mass.
- (D) G-forces acting on an astronaut increase as the craft re-enters the atmosphere.

14. A negatively charged particle enters a region with a uniform electric field. The direction of the particle's velocity and the electric field are shown in the diagram

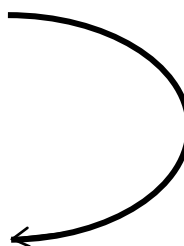


Which diagram best represents the subsequent path of the particle in the magnetic field?

(A)



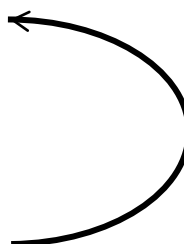
(B)



(C)



(D)



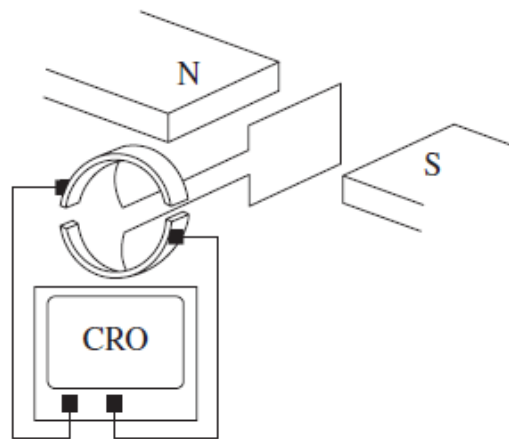
15. A photon of light has a wavelength of 580nm. What is its energy in Joules?

- (A) 3.43×10^{-19}
- (B) 3.43×10^{-28}
- (C) 3.84×10^{-40}
- (D) 3.84×10^{-31}

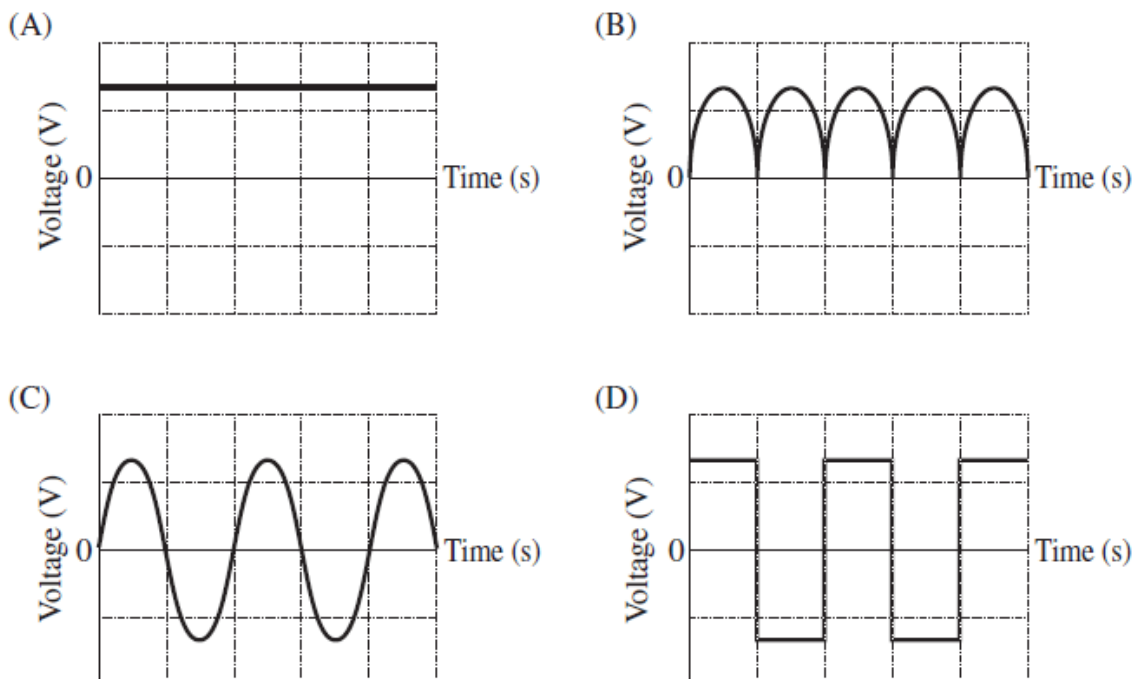
16. Why is high voltage used to transmit electrical energy from the power station to users?

- (A) High voltages result in lower currents which are safer.
- (B) It allows supporting structures to have smaller insulators.
- (C) It minimises the effects of the electrical resistance of the wires.
- (D) It ensures that, even with voltage losses that 240V still reaches the user.

17. The diagram shows a DC generator connected to a cathode ray oscilloscope(CRO).

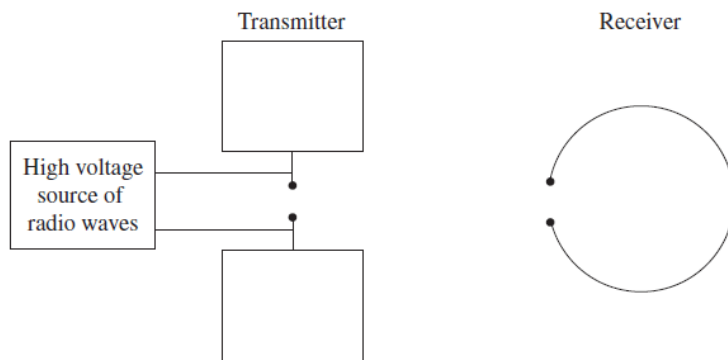


Which response best shows the voltage signal displayed on the oscilloscope?



Use the following information to answer questions 18 and 19.

Heinrich Hertz used a set-up similar to the one shown below to measure the speed of radio waves.



18. What vital piece of equipment is missing from the apparatus shown?

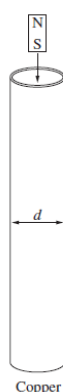
- (A) A screen
- (B) A voltmeter
- (C) A galvanometer
- (D) A metal reflector

19. A glass sheet was placed between the transmitter and receiver.

Which of the following observations is consistent with the photoelectric effect that Hertz produced?

- (A) Radio waves were blocked when the glass sheet was in place.
- (B) Ultraviolet waves were blocked when the glass sheet was in place.
- (C) The maximum spark length was longer when the glass sheet was in place.
- (D) The maximum spark length was shorter when the glass sheet was in place.

20. A bar magnet is dropped through a long copper tube as shown. It was observed that the magnet hit the ground in 5 seconds.

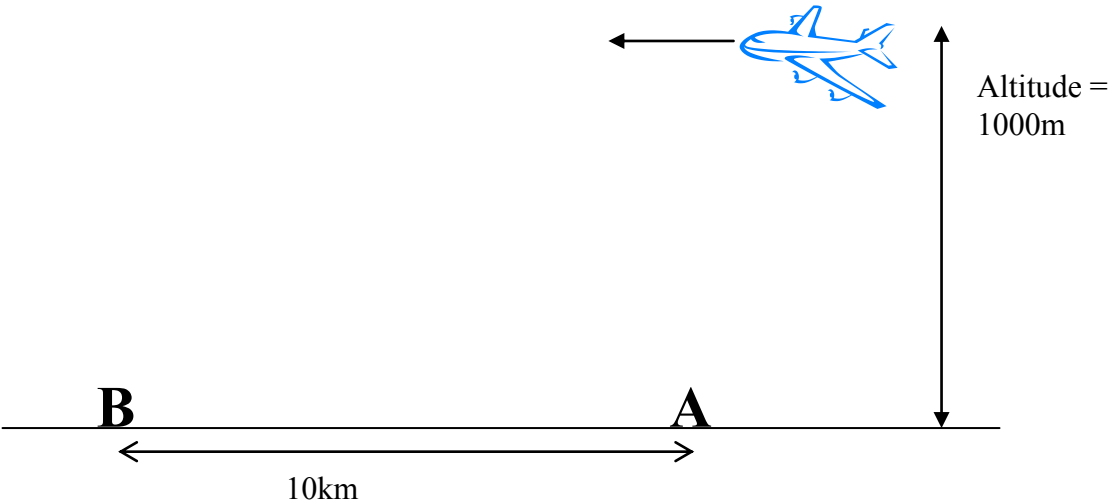


Which modification to the experiment will most significantly decrease the time it takes the magnet to hit the ground?

- (A) Cut vertical laminations in the tube.
- (B) Double the diameter of the tube.
- (C) Use a plastic tube instead of a copper tube.
- (D) Propel the magnet downwards with an initial velocity of 5ms^{-1} .

Section I – Part B

21. A bomber travels at 200 km/hr horizontally above its desired target as shown below. The bomber needs to hit 2 targets which are a distance of 10km apart. The targets are labelled A and B as follows. If the first bomb is released at exactly 1:00pm and hits target A, then at what time must the second bomb be released in order to hit target B? (4 marks)



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22. (a) With reference to one device studied in each of the core units of the course, identify the major energy transformation that occurs in each device. (3 marks)

Core Unit	Device	Energy Transformation
Space		
Motors and Generators		
Ideas to Implementation		

(b) Account for Lenz’s law in terms of the law of conservation of energy. (3 marks)

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23. (a) Recount the method used in the Michelson-Morley experiment. (4 marks)

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- (b) A student concludes that the Michelson-Morley experiment was a waste of time because they did not achieve the aim of the experiment. Assess the conclusion made by the student. (3 marks)

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24. You carried out an investigation of the striation patterns in a discharge tube.

- (a) Identify the independent and dependent variables and two variables that were kept constant in this investigation. (2 marks)

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- (b) Outline a safety precaution necessary when conducting investigations that use an induction coil. (1 mark)

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25. Triplets (Dave, Bill and Tim) are born in the 22nd century. Dave and Bill are astronauts and both head off on a mission in opposite directions. Tim stays behind on Earth. Dave travels away from the Earth at $0.9c$ relative to Tim. Bill travels in the opposite direction at $0.95c$, also relative to Tim.

(a) Of the triplets who will Tim observe to be ageing most quickly? (1 mark)

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(b) From Bill's perspective it takes him 1 month to reach his destination. How long will Tim observe Bill's trip to take? (2 marks)

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(c) A local newspaper reports that Dave and Bill are moving away from each other at 1.85 times the speed of light. Assess this statement. (3 marks)

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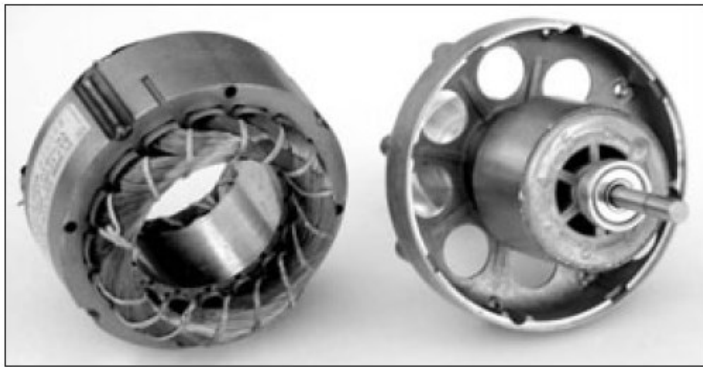
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26. The following shows a picture of an AC induction motor.



(a) Compare the structure of this motor with a synchronous AC motor. (4 marks)

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(b) Recount the experiment that you conducted in class to demonstrate the principal by which the AC induction motor works. (3 marks)

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27. Three long, straight, parallel wires; X, Y and Z are positioned as shown below. Each wire is 1m long and the current in wire Z runs in the opposite direction to both wire X and Y.



Calculate the force on wire Y, due to the other two wires. **(3 marks)**

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28. JJ Thomson conducted an investigation to measure the charge to mass ratio of cathode rays. In one part of his experiment he ensured that the cathode ray passed through the discharge tube undeflected. Describe how he was able to do this and explain why this enabled him to measure the velocity of the cathode rays he was observing. **(5 marks)**

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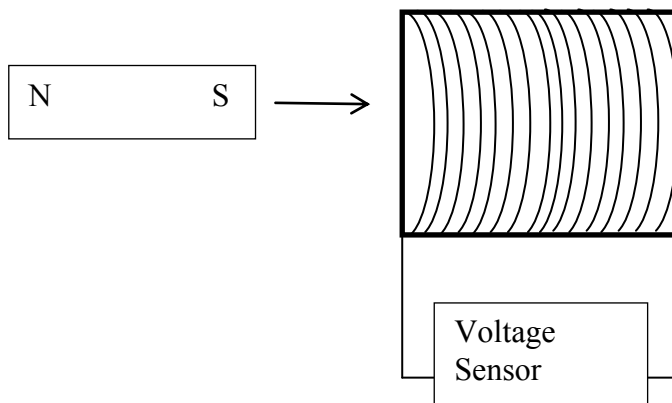
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29. An astronaut in the international space station (satellite) conducts an investigation. During this investigation a magnet is allowed to pass through a solenoid connected to a voltage sensor and data logger. As the magnet approaches the solenoid it has a constant velocity and is not being forced in any way by the astronaut. The magnet passes through the solenoid and out the other side



Sketch a graph of induced voltage vs time for this experiment.(3marks)



30. Analyse the relationship between the work done by Einstein and Planck and the shift in scientific thinking about the nature of light. **(7 marks)**

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31. Explain why the discharge tube pictured below was not able to resolve the debate as to whether cathode rays were charged particles or a form of electromagnetic radiation. **(4 marks)**



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End of Part B

Section II – Medical Physics**25 Marks**

Allow about 45 minutes for this section. Answer all parts of this question in a writing booklet.

Additional writing booklets are available if required.

(a)(i) Describe the Doppler effect (1 mark)

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(ii) Using an example of cardiac problem explain why Doppler ultrasound imaging can provide a diagnosis. (3 marks)

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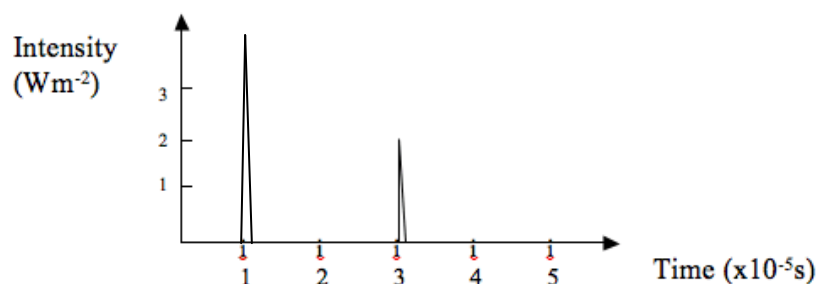
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(b) An A-scan ultrasound is used to assist in making a medical diagnosis. The ultrasound travels through soft body tissue and strikes an organ. Sound is reflected from both the front and back of the organ. Sound travels at an average speed of 1500 ms^{-1} in soft body tissue. The A-scan record below shows the result of this process for ultrasound having an initial intensity 100 W m^{-2} . The initial pulse was applied at $t = 0 \text{ s}$.



Using the information in the graph determine the size of the organ being examined. (3 marks)

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(c)

Body Tissue	Density (kg m^{-3})	Velocity (m s^{-1})
Blood	1060	1570
Average Soft body	1000	1500
Brain	1025	1540
Fat	925	1450
Muscle	1075	1590

(i) Calculate the acoustic impedance of fat tissue. (1 mark)

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(ii) The acoustic impedance of air is $400 \text{ kg m}^{-2} \text{ s}^{-1}$. Using a calculation explain why ultrasound is not an effective tool in imaging the lungs.

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(d) Discuss the use of the diagnostic techniques used to produce the following images.(4 marks)



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- (e) Many studies have found a link between CAT scans and an increased risk of brain cancers. Considering this risk, assess the impact of this type of imaging on society. **(5 marks)**

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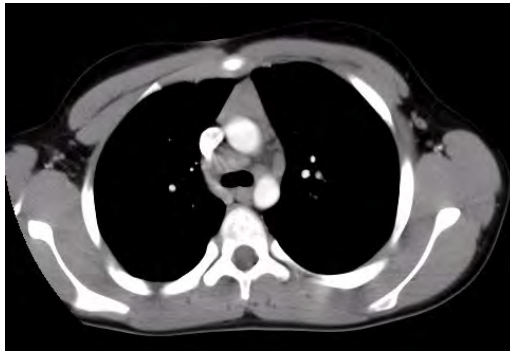
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- (f) The following image is a CAT scan of the chest



- (i) Describe the features of this image. (2 marks)

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- (ii) Outline the process used to produce the image. (3 marks)

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END OF EXAM

HIGHER SCHOOL CERTIFICATE EXAMINATION

Physics

DATA SHEET

Charge on electron, q_e	$-1.602 \times 10^{-19} \text{ C}$
Mass of electron, m_e	$9.109 \times 10^{-31} \text{ kg}$
Mass of neutron, m_n	$1.675 \times 10^{-27} \text{ kg}$
Mass of proton, m_p	$1.673 \times 10^{-27} \text{ kg}$
Speed of sound in air	340 m s^{-1}
Earth's gravitational acceleration, g	9.8 m s^{-2}
Speed of light, c	$3.00 \times 10^8 \text{ m s}^{-1}$
Magnetic force constant, $\left(k \equiv \frac{\mu_0}{2\pi}\right)$	$2.0 \times 10^{-7} \text{ N A}^{-2}$
Universal gravitational constant, G	$6.67 \times 10^{-11} \text{ N m}^2 \text{ kg}^{-2}$
Mass of Earth	$6.0 \times 10^{24} \text{ kg}$
Planck constant, h	$6.626 \times 10^{-34} \text{ J s}$
Rydberg constant, R (hydrogen)	$1.097 \times 10^7 \text{ m}^{-1}$
Atomic mass unit, u	$1.661 \times 10^{-27} \text{ kg}$ $931.5 \text{ MeV}/c^2$
1 eV	$1.602 \times 10^{-19} \text{ J}$
Density of water, ρ	$1.00 \times 10^3 \text{ kg m}^{-3}$
Specific heat capacity of water	$4.18 \times 10^3 \text{ J kg}^{-1} \text{ K}^{-1}$

FORMULAE SHEET

$$v = f\lambda$$

$$I \propto \frac{1}{d^2}$$

$$\frac{v_1}{v_2} = \frac{\sin i}{\sin r}$$

$$E = \frac{F}{q}$$

$$R = \frac{V}{I}$$

$$P = VI$$

$$\text{Energy} = VIt$$

$$v_{\text{av}} = \frac{\Delta r}{\Delta t}$$

$$a_{\text{av}} = \frac{\Delta v}{\Delta t} \text{ therefore } a_{\text{av}} = \frac{v - u}{t}$$

$$\Sigma F = ma$$

$$F = \frac{mv^2}{r}$$

$$E_k = \frac{1}{2}mv^2$$

$$W = Fs$$

$$p = mv$$

$$\text{Impulse} = Ft$$

$$E_p = -G \frac{m_1 m_2}{r}$$

$$F = mg$$

$$v_x^2 = u_x^2$$

$$v = u + at$$

$$v_y^2 = u_y^2 + 2a_y \Delta y$$

$$\Delta x = u_x t$$

$$\Delta y = u_y t + \frac{1}{2}a_y t^2$$

$$\frac{r^3}{T^2} = \frac{GM}{4\pi^2}$$

$$F = \frac{Gm_1 m_2}{d^2}$$

$$E = mc^2$$

$$l_v = l_0 \sqrt{1 - \frac{v^2}{c^2}}$$

$$t_v = \frac{t_0}{\sqrt{1 - \frac{v^2}{c^2}}}$$

$$m_v = \frac{m_0}{\sqrt{1 - \frac{v^2}{c^2}}}$$

FORMULAE SHEET

$$\frac{F}{l} = k \frac{I_1 I_2}{d}$$

$$d = \frac{1}{p}$$

$$F = BIl \sin \theta$$

$$M = m - 5 \log \left(\frac{d}{10} \right)$$

$$\tau = Fd$$

$$\frac{I_A}{I_B} = 100^{(m_B - m_A)/5}$$

$$\tau = nBIA \cos \theta$$

$$m_1 + m_2 = \frac{4\pi^2 r^3}{GT^2}$$

$$\frac{V_p}{V_s} = \frac{n_p}{n_s}$$

$$F = qvB \sin \theta$$

$$\frac{1}{\lambda} = R \left(\frac{1}{n_f^2} - \frac{1}{n_i^2} \right)$$

$$E = \frac{V}{d}$$

$$\lambda = \frac{h}{mv}$$

$$E = hf$$

$$c = f\lambda$$

$$A_0 = \frac{V_{\text{out}}}{V_{\text{in}}}$$

$$Z = \rho v$$

$$\frac{V_{\text{out}}}{V_{\text{in}}} = -\frac{R_f}{R_i}$$

$$\frac{I_r}{I_0} = \frac{[Z_2 - Z_1]^2}{[Z_2 + Z_1]^2}$$

PERIODIC TABLE OF THE ELEMENTS

		KEY															
Atomic Number	Symbol of element	Atomic Number	Symbol of element	Atomic Number	Symbol of element	Atomic Number	Symbol of element	Atomic Number	Symbol of element	Atomic Number	Symbol of element	Atomic Number	Symbol of element	Atomic Number	Symbol of element	Atomic Number	Symbol of element
1 H 1.008 Hydrogen		79 Au 197.0 Gold		26 Fe 55.85 Iron		27 Co 58.93 Cobalt		28 Ni 58.69 Nickel		29 Cu 63.55 Copper		30 Zn 65.41 Zinc		31 Ga 69.72 Gallium		32 Ge 72.64 Germanium	
3 Li 6.941 Lithium	4 Be 9.012 Beryllium			25 Mn 54.94 Manganese		24 Cr 52.00 Chromium		23 V 50.94 Vanadium		22 Ti 47.87 Titanium		21 Sc 44.96 Scandium		20 Ca 40.08 Calcium		19 K 39.10 Potassium	
11 Na 22.99 Sodium	12 Mg 24.31 Magnesium			43 Tc [97.91] Technetium		42 Mo 95.94 Molybdenum		41 Nb 92.91 Niobium		40 Zr 91.22 Zirconium		39 Y 88.91 Yttrium		38 Sr 87.62 Strontium		37 Rb 85.47 Rubidium	
55 Cs 132.9 Caesium	56 Ba 137.3 Barium			75 Re 186.2 Rhenium		74 W 183.8 Tungsten		73 Ta 180.9 Tantalum		72 Hf 178.5 Hafnium		71 La 138.9 Lanthanum		70 Yb 173.0 Ytterbium		69 Tm 168.9 Thulium	
87 Fr [223] Francium	88 Ra [226] Radium			107 Bh [264] Bohrium		106 Sg [266] Seaborgium		105 Db [262] Dubnium		104 Rf [261] Rutherfordium		103 Lr [260] Lawrencium		102 No [259] Nobelium		101 Md [258] Mendelevium	
				108 Hs [277] Hassium		109 Mt [268] Meitnerium		110 Ds [271] Darmstadtium		111 Rg [272] Roentgenium		112 Cn [285] Copernicium		113 Nh [286] Nihonium		114 Fl [289] Flerovium	
				150 Uu [384] Ununnilium		149 Uub [385] Ununnilium		148 Uuh [386] Ununnilium		147 Uuq [387] Ununnilium		146 Uuq [387] Ununnilium		145 Uup [387] Ununnilium		144 Uuq [387] Ununnilium	
				165 Uuh [489] Ununhexium		164 Uuq [489] Ununnilium		163 Uuh [489] Ununhexium		162 Uuq [489] Ununnilium		161 Uuh [489] Ununhexium		160 Uuq [489] Ununnilium		159 Uuh [489] Ununhexium	
				173 Uuh [583] Ununseptium		172 Uuq [583] Ununnilium		171 Uuh [583] Ununseptium		170 Uuq [583] Ununnilium		169 Uuh [583] Ununseptium		168 Uuq [583] Ununnilium		167 Uuh [583] Ununseptium	
				188 Uuh [709] Ununoctium		187 Uuq [709] Ununnilium		186 Uuh [709] Ununoctium		185 Uuq [709] Ununnilium		184 Uuh [709] Ununoctium		183 Uuq [709] Ununnilium		182 Uuh [709] Ununoctium	
				201 Uuh [843] Unbihexium		200 Uuq [843] Ununnilium		199 Uuh [843] Unbihexium		198 Uuq [843] Ununnilium		197 Uuh [843] Unbihexium		196 Uuq [843] Ununnilium		195 Uuh [843] Unbihexium	
				210 Uuh [843] Unbihexium		209 Uuq [843] Ununnilium		208 Uuh [843] Unbihexium		207 Uuq [843] Ununnilium		206 Uuh [843] Unbihexium		205 Uuq [843] Ununnilium		204 Uuh [843] Unbihexium	
				223 Uuh [1049] Untrihexium		222 Uuq [1049] Ununnilium		221 Uuh [1049] Untrihexium		220 Uuq [1049] Ununnilium		219 Uuh [1049] Untrihexium		218 Uuq [1049] Ununnilium		217 Uuh [1049] Untrihexium	
				238 Uuh [226] Untrihexium		237 Uuq [226] Ununnilium		236 Uuh [226] Untrihexium		235 Uuq [226] Ununnilium		234 Uuh [226] Untrihexium		233 Uuq [226] Ununnilium		232 Uuh [226] Untrihexium	
				252 Uuh [286] Untrihexium		251 Uuq [286] Ununnilium		250 Uuh [286] Untrihexium		249 Uuq [286] Ununnilium		248 Uuh [286] Untrihexium		247 Uuq [286] Ununnilium		246 Uuh [286] Untrihexium	
				261 Uuh [304] Untrihexium		260 Uuq [304] Ununnilium		259 Uuh [304] Untrihexium		258 Uuq [304] Ununnilium		257 Uuh [304] Untrihexium		256 Uuq [304] Ununnilium		255 Uuh [304] Untrihexium	
				271 Uuh [349] Untrihexium		270 Uuq [349] Ununnilium		269 Uuh [349] Untrihexium		268 Uuq [349] Ununnilium		267 Uuh [349] Untrihexium		266 Uuq [349] Ununnilium		265 Uuh [349] Untrihexium	
				285 Uuh [415] Untrihexium		284 Uuq [415] Ununnilium		283 Uuh [415] Untrihexium		282 Uuq [415] Ununnilium		281 Uuh [415] Untrihexium		280 Uuq [415] Ununnilium		279 Uuh [415] Untrihexium	
				294 Uuh [489] Untrihexium		293 Uuq [489] Ununnilium		292 Uuh [489] Untrihexium		291 Uuq [489] Ununnilium		290 Uuh [489] Untrihexium		289 Uuq [489] Ununnilium		288 Uuh [489] Untrihexium	
				304 Uuh [564] Untrihexium		303 Uuq [564] Ununnilium		302 Uuh [564] Untrihexium		301 Uuq [564] Ununnilium		300 Uuh [564] Untrihexium		299 Uuq [564] Ununnilium		298 Uuh [564] Untrihexium	
				312 Uuh [600] Untrihexium		311 Uuq [600] Ununnilium		310 Uuh [600] Untrihexium		309 Uuq [600] Ununnilium		308 Uuh [600] Untrihexium		307 Uuq [600] Ununnilium		306 Uuh [600] Untrihexium	
				315 Uuh [600] Untrihexium		314 Uuq [600] Ununnilium		313 Uuh [600] Untrihexium		312 Uuq [600] Ununnilium		311 Uuh [600] Untrihexium		310 Uuq [600] Ununnilium		309 Uuh [600] Untrihexium	
				318 Uuh [600] Untrihexium		317 Uuq [600] Ununnilium		316 Uuh [600] Untrihexium		315 Uuq [600] Ununnilium		314 Uuh [600] Untrihexium		313 Uuq [600] Ununnilium		312 Uuh [600] Untrihexium	
				329 Uuh [709] Untrihexium		328 Uuq [709] Ununnilium		327 Uuh [709] Untrihexium		326 Uuq [709] Ununnilium		325 Uuh [709] Untrihexium		324 Uuq [709] Ununnilium		323 Uuh [709] Untrihexium	
				344 Uuh [843] Untrihexium		343 Uuq [843] Ununnilium		342 Uuh [843] Untrihexium		341 Uuq [843] Ununnilium		340 Uuh [843] Untrihexium		339 Uuq [843] Ununnilium		338 Uuh [843] Untrihexium	
				354 Uuh [900] Untrihexium		353 Uuq [900] Ununnilium		352 Uuh [900] Untrihexium		351 Uuq [900] Ununnilium		350 Uuh [900] Untrihexium		349 Uuq [900] Ununnilium		348 Uuh [900] Untrihexium	
				364 Uuh [969] Untrihexium		363 Uuq [969] Ununnilium		362 Uuh [969] Untrihexium		361 Uuq [969] Ununnilium		360 Uuh [969] Untrihexium		359 Uuq [969] Ununnilium		358 Uuh [969] Untrihexium	
				374 Uuh [1037] Untrihexium		373 Uuq [1037] Ununnilium		372 Uuh [1037] Untrihexium		371 Uuq [1037] Ununnilium		370 Uuh [1037] Untrihexium		369 Uuq [1037] Ununnilium		368 Uuh [1037] Untrihexium	
				384 Uuh [1115] Untrihexium		383 Uuq [1115] Ununnilium		382 Uuh [1115] Untrihexium		381 Uuq [1115] Ununnilium		380 Uuh [1115] Untrihexium		379 Uuq [1115] Ununnilium		378 Uuh [1115] Untrihexium	
				396 Uuh [1200] Untrihexium		395 Uuq [1200] Ununnilium		394 Uuh [1200] Untrihexium		393 Uuq [1200] Ununnilium		392 Uuh [1200] Untrihexium		391 Uuq [1200] Ununnilium		390 Uuh [1200] Untrihexium	
				404 Uuh [1276] Untrihexium		403 Uuq [1276] Ununnilium		402 Uuh [1276] Untrihexium		401 Uuq [1276] Ununnilium		400 Uuh [1276] Untrihexium		399 Uuq [1276] Ununnilium		398 Uuh [1276] Untrihexium	
				412 Uuh [1320] Untrihexium		411 Uuq [1320] Ununnilium		410 Uuh [1320] Untrihexium		409 Uuq [1320] Ununnilium		408 Uuh [1320] Untrihexium		407 Uuq [1320] Ununnilium		406 Uuh [1320] Untrihexium	
				422 Uuh [1409] Untrihexium		421 Uuq [1409] Ununnilium		420 Uuh [1409] Untrihexium		419 Uuq [1409] Ununnilium		418 Uuh [1409] Untrihexium		417 Uuq [1409] Ununnilium		416 Uuh [1409] Untrihexium	
				432 Uuh [1500] Untrihexium		431 Uuq [1500] Ununnilium		430 Uuh [1500] Untrihexium		429 Uuq [1500] Ununnilium		428 Uuh [1500] Untrihexium		427 Uuq [1500] Ununnilium		426 Uuh [1500] Untrihexium	
				442 Uuh [1599] Untrihexium		441 Uuq [1599] Ununnilium		440 Uuh [1599] Untrihexium		439 Uuq [1599] Ununnilium		438 Uuh [1599] Untrihexium		437 Uuq [1599] Ununnilium		436 Uuh [1599] Untrihexium	
				452 Uuh [1697] Untrihexium		451 Uuq [1697] Ununnilium		450 Uuh [1697] Untrihexium		449 Uuq [1697] Ununnilium		448 Uuh [1697] Untrihexium		447 Uuq [1697] Ununnilium		446 Uuh [1697] Untrihexium	
				462 Uuh [1799] Untrihexium		461 Uuq [1799] Ununnilium		460 Uuh [1799] Untrihexium		459 Uuq [1799] Ununnilium		458 Uuh [1799] Untrihexium		457 Uuq [1799] Ununnilium		456 Uuh [1799] Untrihexium	
				472 Uuh [1903] Untrihexium		471 Uuq [1903] Ununnilium		470 Uuh [1903] Untrihexium		469 Uuq [1903] Ununnilium		468 Uuh [1903] Untrihexium		467 Uuq [1903] Ununnilium		466 Uuh [1903] Untrihexium	
				482 Uuh [2011] Untrihexium		481 Uuq [2011] Ununnilium		480 Uuh [2011] Untrihexium		479 Uuq [2011] Ununnilium		478 Uuh [2011] Untrihexium		477 Uuq [2011] Ununnilium		476 Uuh [2011] Untrihexium	
				492 Uuh [2121] Untrihexium		491 Uuq [2121] Ununnilium		490 Uuh [2121] Untrihexium		489 Uuq [2121] Ununnilium		488 Uuh [2121] Untrihexium		487 Uuq [2121] Ununnilium		486 Uuh [2121] Untrihexium	
				502 Uuh [2233] Untrihexium		501 Uuq [2233] Ununnilium		500 Uuh [2233] Untrihexium		499 Uuq [2233] Ununnilium		498 Uuh [2233] Untrihexium		497 Uuq [2233] Ununnilium		496 Uuh [2233] Untrihexium	
				512 Uuh [2345] Untrihexium		511 Uuq [2345] Ununnilium		510 Uuh [2345] Untrihexium		509 Uuq [2345] Ununnilium		508 Uuh [2345] Untrihexium		507 Uuq [2345] Ununnilium		506 Uuh [2345] Untrihexium	
				522 Uuh [2459] Untrihexium		521 Uuq [2459] Ununnilium		520 Uuh [2459] Untrihexium		519 Uuq [2459] Ununnilium		518 Uuh [2459] Untrihexium		517 Uuq [2459] Ununnilium		516 Uuh [2459] Untrihexium	
				532 Uuh [2577] Untrihexium		531 Uuq [2577] Ununnilium		530 Uuh [2577] Untrihexium		529 Uuq [2577] Ununnilium		528 Uuh [2577] Untrihexium		527 Uuq [2577] Ununnilium		526 Uuh [2577] Untrihexium	
				542 Uuh [2699] Untrihexium		541 Uuq [2699] Ununnilium		540 Uuh [2699] Untrihexium		539 Uuq [2699] Ununnilium		538 Uuh [2699] Untrihexium		537 Uuq [2699] Ununnilium		536 Uuh [2699] Untrihexium	
				552 Uuh [2823] Untrihexium		551 Uuq [2823] Ununnilium		550 Uuh [2823] Untrihexium		549 Uuq [2823] Ununnilium		548 Uuh [2823] Untrihexium		547 Uuq [2823] Ununnilium		546 Uuh [2823] Untrihexium	
				562 Uuh [2951] Untrihexium		561 Uuq [2951] Ununnilium		560 Uuh [2951] Untrihexium		559 Uuq [2951] Ununnilium		558 Uuh [2951] Untrihexium		557 Uuq [2951] Ununnilium		556 Uuh [2951] Untrihexium	
				572 Uuh [3083] Untrihexium		571 Uuq [3083] Ununnilium		570 Uuh [3083] Untrihexium		569 Uuq [3083] Ununnilium		568 Uuh [3083] Untrihexium		567 Uuq [3083] Ununnilium		566 Uuh [3083] Untrihexium	
				582 Uuh [3219] Untrihexium		581 Uuq [3219] Ununnilium		580 Uuh [3219] Untrihexium		579 Uuq [3219] Ununnilium		578 Uuh [3219] Untrihexium		577 Uuq [3219] Ununnilium		576 Uuh [3219] Untrihexium	
				592 Uuh [3359] Untrihexium		591 Uuq [3359] Ununnilium		590 Uuh [3359] Untrihexium		589 Uuq [3359] Ununnilium		588 Uuh [3359] Untrihexium		587 Uuq [3359] Ununnilium		586 Uuh [3359] Untrihexium	
				602 Uuh [3503] Untrihexium		601 Uuq [3503] Ununnilium		600 Uuh [3503] Untrihexium		599 Uuq [3							

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Multiple-choice Answer Sheet

Select the alternative A, B, C or D that best answers the question. Fill in the response oval completely.

Sample $2 + 4 =$ (A) 2 (B) 6 (C) 8 (D) 9

(A) ☐ (B) ☒ (C) ☐ (D) ☐

If you think you have made a mistake, put a cross through the incorrect answer and fill in the new answer.

(A) ☒ (B) ☒ (C) ☐ (D) ☐

If you change your mind and have crossed out what you consider to be the correct answer, then indicate this by writing the word correct and drawing an arrow as follows:

correct
↓

(A) ☒ (B) ☒ (C) ☐ (D) ☐

Part A

- | | | | | |
|-----|---------------------------|---------------------------|---------------------------|---------------------------|
| 1. | (A) ☐ | (B) ☐ | (C) ☐ | (D) ☐ |
| 2. | (A) ☐ | (B) ☐ | (C) ☐ | (D) ☐ |
| 3. | (A) ☐ | (B) ☐ | (C) ☐ | (D) ☐ |
| 4. | (A) ☐ | (B) ☐ | (C) ☐ | (D) ☐ |
| 5. | (A) ☐ | (B) ☐ | (C) ☐ | (D) ☐ |
| 6. | (A) ☐ | (B) ☐ | (C) ☐ | (D) ☐ |
| 7. | (A) ☐ | (B) ☐ | (C) ☐ | (D) ☐ |
| 8. | (A) ☐ | (B) ☐ | (C) ☐ | (D) ☐ |
| 9. | (A) ☐ | (B) ☐ | (C) ☐ | (D) ☐ |
| 10. | (A) ☐ | (B) ☐ | (C) ☐ | (D) ☐ |
| 11. | (A) ☐ | (B) ☐ | (C) ☐ | (D) ☐ |
| 12. | (A) ☐ | (B) ☐ | (C) ☐ | (D) ☐ |
| 13. | (A) ☐ | (B) ☐ | (C) ☐ | (D) ☐ |
| 14. | (A) ☐ | (B) ☐ | (C) ☐ | (D) ☐ |
| 15. | (A) ☐ | (B) ☐ | (C) ☐ | (D) ☐ |
| 16. | (A) ☐ | (B) ☐ | (C) ☐ | (D) ☐ |
| 17. | (A) ☐ | (B) ☐ | (C) ☐ | (D) ☐ |
| 18. | (A) ☐ | (B) ☐ | (C) ☐ | (D) ☐ |
| 19. | (A) ☐ | (B) ☐ | (C) ☐ | (D) ☐ |
| 20. | (A) ☐ | (B) ☐ | (C) ☐ | (D) ☐ |

Marking Criteria

M/C: 1 B 2 C 3 A 4(na) 5C 6D 7D 8B 9B 10B 11A 12B 13A 14B 15B 16C 17B 18D 19D 20C

21. Criteria	Mark/s
Correctly calculates 1:03pm as time for bomb release	4
Correctly calculates time for plane to travel 10 km.	3
Incorrect time for bomb release but correct calculation of time for plane to travel 10 km and time for bomb to hit the ground	2
Calculates time for bomb to reach the ground, or use of $v=r/t$.	1

22.(a) Criteria	Mark/s
Identifies three significant devices we have studied and their major energy transformations	3
Identifies two significant devices we have studied and their major energy transformations	2
Identifies one significant device we have studied and its major energy transformation	1

22.(b) Criteria	Mark/s
Answer demonstrates an understanding of both the law of conservation of energy and Lenz's law. Answer correctly relates loss in kinetic energy with increase in electrical energy, linking the law of conservation of energy and Lenz's law.	3
Answer demonstrates an understanding of both the law of conservation of energy and Lenz's law	2
Answer demonstrates an understanding of either the law of conservation of energy and Lenz's law	1

23. (a) Criteria	Mark/s
Correct diagram of Michelson-Morley experiment. Includes in answer the observation of an interference pattern and rotation of the apparatus through 90°	4
Correct answer missing either observation of an interference pattern or rotation of the apparatus through 90° or one mistake in the diagram	3
Correct answer including two errors in either the diagram or answer	2
Correct answer including three errors in either the diagram or answer	1

23b. Criteria	Mark/s
Makes a judgement of the statement and justifies by relating the null result to later development is scientific thinking such as Einstein's postulate regarding the constancy of the speed of light.	3
Makes a judgement of the statement and states the results and makes some reference to changes in scientific thinking.	2
Demonstrates a limited understanding of the importance of the results of this experiment.	1

24(a) Criteria	Mark/s
Correctly identifies the independent variable (pressure) and the dependent variable (striation pattern) AND Two significant controlled variables (eg voltage, gas in tube (air), length of tubes.	2
Correctly identifies the independent variable and the dependent variable OR Two significant controlled variables	1

24(b) Criteria	Mark/s
One significant safety precaution related to the induction coil	1

25(a) Criteria	Mark/s
Tim	1

25(b) Criteria	Mark/s
Correct answer of 3.2 months	2
Correct substitution into equation with incorrect answer	1

25. Criteria	Mark/s
Judgement of statement with significant	3
	2
	1

26. Criteria	Mark/s
Judgement of importance AND Significant reasoning related to safety and efficiency AND Demonstrates a logical progression and good use of scientific terminology	6
Judgement of importance AND Significant reasoning related to safety and efficiency	4-5
Limited discussion of issues related to safety and efficiency	2-3
One correct point related to efficiency or safety.	1

27. Criteria	Mark/s
Answer calculates the force per unit length due to wire X and Y and adds them using appropriate vector addition.	3
Answer calculates the force per unit length due to wire X and Y and adds them incorrectly. OR Correct method with one error or substitution..	2
Correct method with two errors	1

28. Criteria	Mark/s
Answer explains how the Earth's orbital motion and the slingshot effect can be utilised to minimise the fuel needed by the probe.	4
Answer demonstrates some understanding of how orbital motion of the Earth and the slingshot effect are utilised.	2-3
Answer identifies that the slingshot effect or orbital motion of the Earth is utilised.	1

29. Criteria	Mark/s
Explanation of why increasing voltage decreases current and relates the change in current to the wires in the secondary being thinner.	3
Explanation of why increasing voltage decreases current.	2
Identifies that increasing voltage decreases current.	1

30. Criteria	Mark/s
Correct description of method including - A suitable method of controlling speed. - The use of a galvanometer or micro ammeter - Measurements across a specific range of distances. - Repetition of each measurement at least three times.	4
Correct description of method with one of the above omitted.	3
Correct description of method with two of the above omitted.	2
Correct description of method with three of the above omitted.	1

31 Criteria	Mark/s
Well written response which includes - Identification of the tube (maltese cross) - A specific observation that the cathode rays formed a shadow and this means that - Cathode rays travel in straight lines but - This inference can be explained using both the EM wave and particle models proposed for cathode rays. (including an outline of how the two models explain the observation(s)) - And therefore this tube could not resolve the debate	3-4

Identifies the type of tube and the relevant observation but does not clearly link this to the fact that both models can explain this behaviour OR Response includes information based on observations/inferences from other CRT's (eg the paddle wheel) in addition to the points raised above.	1-2
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Section II – Medical Physics

A (i) Criteria	Mark/s
Describes the Doppler effect as the observed change in frequency of a wave (or sound) due to the relative motion of wave source and the observer.	1

a (ii). Criteria	Mark/s
Coherent response which includes - a specific cardiac problem (eg. faulty heart valve, blocked coronary artery, hole in the wall between ventricles) - an outline of how Doppler ultrasound can image movement of blood and - clearly relates this technique to the identified cardiac problem	3
Response includes the above information but lacks important detail (eg identified medical problem not specific to the heart OR does not make a specific link between the technique and the medical problem) or Response contains one significant factual error	2
Response demonstrates a basic understanding of how Doppler ultrasound can be used to image moving tissue	1

b. Criteria	Mark/s
Correctly identifies the time interval between the two peaks on the graph (2×10^{-5} s) and Substitutes correctly into the formula $v = r/t$ to calculate the distance travelled by the ultrasound in the organ (0.03 m) and Recognises that this is twice the size of the organ – final answer is 0.015 m or 1.5 cm	3
Correctly calculates the distance travelled by the ultrasound in the organ	2
Correctly identifies the time interval between the two peaks on the graph	1

c (i). Criteria	Mark/s
Substitutes correctly into the equation $Z = \rho v$ to determine the acoustic impedance ($Z = 1341250 \text{ kg m}^{-2} \text{ s}^{-1}$)	1

c (ii). Criteria	Mark/s
Correctly calculates the percentage of sound reflected at a soft tissue/air boundary (99.89 %) and Clearly relates the calculation to the structure of the lungs and States that the intensity of sound penetrating the lungs is too low to produce an image of the internal structure	3
Correctly calculates the percentage of sound reflected at a soft tissue/air boundary and Relates the calculation to the strong reflection of sound from the lungs without linking it specifically to lung structure.	2
Correctly calculates the percentage of sound reflected at a soft tissue/air boundary OR Shows a basic understanding of the importance of a large difference in acoustic impedance in imaging the lungs	1

d. Criteria	Mark/s
Coherent and logical discussion which includes - A correct identification of the two techniques (conventional X-ray and ultrasound sector scan) - A relevant use of each technique outlined (eg X-rays used to identify bone fractures, US used to diagnose health of developing embryo) - A significant point for and against each technique in relation to the identified use	3-4

Response correctly identifies the two techniques and outlines a significant point for OR a significant point against for each.	1-2
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e Criteria	Mark/s
Response makes a clear judgement of the size of the impact on society and Demonstrates a thorough knowledge of CAT scans including - Identification of at least two different conditions that can be identified using CAT scans - At least two relevant features of CAT scans that relate to their ability to detect these conditions (eg high resolution, 2 D to show position of structures, expense) and Clearly links the identified uses of CAT scans to their impacts on society (eg improved knowledge of the anatomy of the human body which leads to better health provisions for society, pressure on the health budget to purchase and maintain expensive CAT scanners, early diagnosis contributes to a healthier society) Demonstrates coherence and logical progression and includes correct use of scientific principles and ideas	4-5
Response makes a clear judgement of the size of the impact on society and demonstrates a sound knowledge of CAT scans and clearly links an identified use of CAT scans to a significant impact on society	3
Response demonstrates some knowledge and understanding of the use and impact of CAT scans on society or the individual.	1-2

f (i) Criteria	Mark/s
Describes two significant features of the image - for example 2 dimensional - High resolution - Shows clear distinction between bone and soft tissue using grey scale	2
Describes one of the above	1

f (ii) . Criteria	Mark/s
A thorough description which includes - a fan-shaped beam of X-rays is passed through the body - The X-ray tube is rotated 180° around the body - a ring of detectors on the opposite side of the body detects variations in the intensity of transmitted X-rays - data analysed by computer (A diagram was useful to illustrate these features)	3
Response contains the above points but lacks detail and/or clarity (eg no diagram)	2
Response missing one essential point	1